MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)
MG3VACSTA200 - Applied Statistical Analysis: Ethical Data Collection, Interpretation and Decision making in Society

Duration: 1.5 Hours Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions. Answer any **seven** Questions
Each question carries **2** marks

1. Define sampling. Name any two sampling methods. [K] [1]
2. Define tabulation of data. [K] [1]
3. What is positive, negative and zero correlation? [U] [1]
4. What is a null hypothesis? [K] [2]
5. Give the null hypothesis for testing the difference between two means in large samples. [U] [2]
6. Define p-value. [K] [2]
7. What is the test statistic for testing proportion in large samples? [U] [2]
8. Define a numeric variable in R. [K] [3]
9. What is R Commander? [K] [3]
10. What function is used to calculate mean in R? [U] [1]

[7x2=14]

Part B

Short Essay Type Questions. Answer any **four** Questions
Each question carries **6** marks

11. Describe the essential steps involved in designing a good questionnaire. [A] [3]
12. Find standard deviation for the following data and interpret the result: 5, 10, 15, 20, 25. [U] [1]
13. Distinguish between parameter and statistic with example. [U] [2]
14. What is a paired t-test for related samples? [U] [2]
15. Explain Yule's coefficient of association with an example. [U] [1]
16. Describe the process of performing a t-test for a single mean using R. [U] [3]

[4x6=24]

Part C

Essay Type Questions. Answer any **one** Question
Each question carries **12** marks

17. Calculate mean, median and mode from the following grouped data: [U] [1]

Class	0-10	10-20	20-30	30-40	40-50
Frequency	5	8	12	10	5

18. The following data relates to marital status and performance in an examination. Examine whether the performance depends on marital status [U] [2]

Marital Status	Performance	
	Good	Bad
Married	60	80
Unmarried	20	40

[1x12= 12]